SITE · SCHEDULE · PLANT · MONEY · RISK

The Project Book

How we would site, schedule, run, fund and de-risk a 100 MW hybrid green hydrogen plant at Paradip, and what the numbers say before anyone pours concrete.

WRITTEN BY

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COURSE

Project Management (PRM)
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Builds on Dossier 1, *Without the Hype*, and Dossier 2, *Where the Money Is*

Every number in this book comes from one of two places: a published source (bracketed and linked, for example [?]), or our Python models, which anyone can rerun from the repository described in Section ??. Figures carry a tag for what kind of number they are (REPORTED, ESTIMATED, PROJECTED) and claims that matter carry a confidence tag (HIGH, MEDIUM, LOW).

VISION

A plant a banker would fund

What we decided, why we decided it, and the one sentence this whole book is trying to prove or disprove.

Dossier 1 asked which machine. Dossier 2 asked where the money is. This book asks the only question a project manager is really paid to answer: **can this thing be built on time, run reliably and paid for, and what has to be true for all three at once?**

We have made our choice. Our plant is a **100 MW hybrid**: 80 MW of pressurised alkaline electrolysers as the workhorse, and 20 MW of PEM electrolysers as the fast-moving part that follows the swings of solar and wind. The PEM slot is also our **option bay**. When its stacks wear out, around year eleven, we can drop in anion-exchange membrane (AEM) stacks instead, if AEM has proven itself by then. We sell hydrogen, not ammonia, to a refinery next door on a 25-year fixed price, because that is the one Indian customer who has already signed contracts of that kind [?, ?]. Ammonia stays as a phase-two option.

Three beliefs carry over from the first two dossiers and become rules for this one.

1. **A project is a promise with a price tag and a date.** We judge every choice by what it does to cost, schedule and the probability of keeping both. A single-point plan is a guess wearing a suit; we plan in ranges and report P50 and P80.
2. **The lender is our toughest stakeholder.** Seventy per cent of the money is debt. If the cash flow cannot cover the debt by a margin of 1.3 times in a bad year, nothing else in this book matters.
3. **Keep the future cheap.** The core of the plant uses proven technology. The new technology (AEM) and the second phase are bought as *options*: small payments today for the right, not the duty, to act later.

> *"A green hydrogen plant is an electricity-buying business with a chemistry set attached. Get the power, the debt and the date right, and the chemistry looks after itself."*

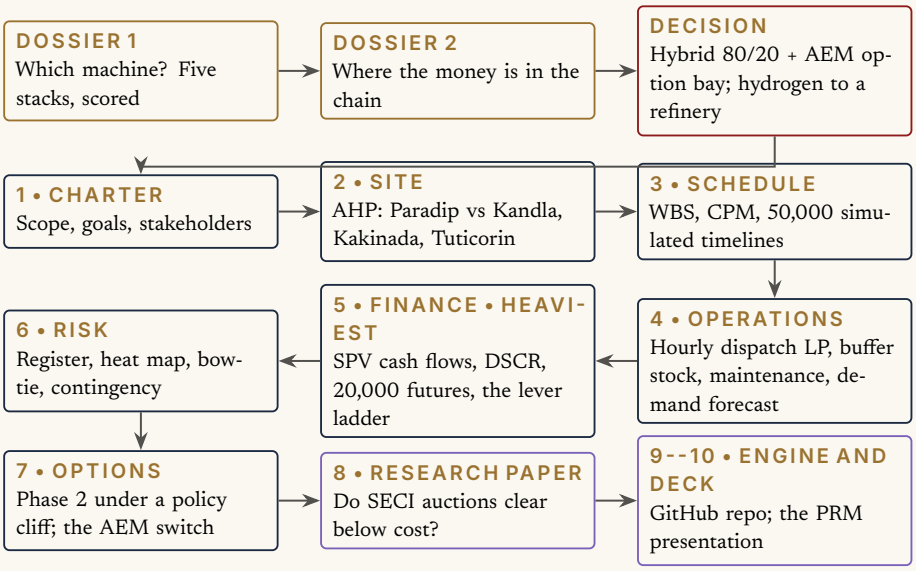
FOUR NUMBERS TO KEEP IN YOUR HEAD

₹522 per kg is what our plant would need to charge, as first designed, to keep its lender and its owners whole. **₹397 per kg** is what the best Indian refinery contract actually pays [?]. **₹389 per kg** is what the plant needs after five design-to-cost moves, so it squeaks through, with ₹8 of headroom. And **57%** is the chance that, even then, the plant breaks its loan covenant at least once in fifteen years. Those four numbers are the story of this book.

WORK FLOW

The map of this book

From a technology choice to a plan a lender, a professor and a reviewer can each test.



The order is deliberate: the site decides which permits sit on the critical path; the schedule decides how much interest piles up before revenue; and only after the finance model can each risk be priced in rupees rather than adjectives.

The decision record

Teams forget why they chose what they chose, so we write it down once, with what would change our minds.

DECISION	WHY	WHAT WOULD CHANGE OUR MIND	CONFIDENCE
Hybrid stack, 80 MW alkaline + 20 MW PEM	Alkaline is the cheapest proven stack (US$3.83/kg); PEM adds fast ramping for about US$0.19/kg more (Section **??**).	Alkaline proven to ramp on hybrid power without extra wear.	MEDIUM
AEM as an option, not a core	Cheap materials, unproven at scale; a listed AEM maker is losing money on returned stacks [?].	40,000+ field hours and a warranty that covers them.	MEDIUM
Sell hydrogen to a refinery	Refineries sign 25-year fixed prices (₹279–397/kg) [?, ?]; ammonia auctions look under-priced (Section **??**).	Ammonia prices that cover cost.	HIGH
Site at Paradip	Buyer next door; port; good grid (Section **??**).	If we sold ammonia or exported, Kandla would win.	MEDIUM

The charter and the people

What we are promising to build, what we are not, and who can help or hurt us along the way.

A project charter is a one-page contract between the team and its sponsor. It is short on purpose: if we cannot say it on one page, we have not decided it yet.

ITEM	WHAT WE COMMIT TO
Project name	**Paradip Green Hydrogen Phase 1** (working title), held in a special-purpose company (SPV) so that its debts do not touch the sponsors' other businesses.
Purpose	Replace about 9,000–9,700 tonnes a year of grey hydrogen at a coastal refinery with certified green hydrogen, at a price fixed for 25 years.
Product and buyer	Hydrogen gas, piped about 5 km to the refinery's hydro-treaters. Price benchmarked on IOCL's Panipat award, ₹397/kg for 25 years [?] REPORTED. The real buyer in our case is a hypothetical refinery at Paradip; IOCL runs one there.
Capacity	100 MW of electrolysis (80 MW alkaline, 20 MW PEM) fed by about 105 MW of solar and 68 MW of wind under contract, a 7-tonne hydrogen buffer, and firm renewable power as back-up (Section ??) ESTIMATED.
Budget	₹1,389 crore as first designed; ₹973 crore is the design-to-cost target that makes the project bankable (Section ??) ESTIMATED.
Schedule	Start January 2027. Most-likely plan 44 months; P50 48 months; P80 51 months (Section ??) ESTIMATED.
Success criteria	(1) Commercial operation by 31 December 2030, the deadline for the 25-year transmission-charge waiver [?]; (2) minimum debt-service cover of at least 1.30 in the lender's base case; (3) equity return of at least 14%; (4) every kilogram certified at or below 2 kg CO_2e per kg of hydrogen under India's Green Hydrogen Standard [?]; (5) zero lost-time injuries.
Sponsor and team	Sponsor: a developer with an investment-grade parent. Project manager: the SPV. For this course: Dhritabrata Swarnakar (finance model, risk, research paper) and Shalvi Bhatnagar (schedule, operations, stakeholders, deck), checking each other's work.

1.1 Scope: what is in, and what is deliberately out

IN SCOPE	OUT OF SCOPE, AND WHY
Electrolysers, rectifiers, transformers, water treatment, compression, buffer storage, 5 km pipeline to the refinery fence	The solar and wind farms themselves: we *buy* their power under contract. Owning them would triple the capital and the risk.
Power purchase agreements and the grid connection	Ammonia synthesis: kept as a phase-two option (Section ??).
Certification, metering and monitoring for the Green Hydrogen Standard	Mobility refuelling: a skid can be bolted on later, but no buyer is big enough to justify it now.
Operations and maintenance for 25 years, including stack replacements	Electrolyser manufacturing: Dossier 2 showed the margins there are thin or negative [?, ?].

1.2 Objectives that can be measured

Each objective names a number and a date, so that at the end we can say "hit" or "missed", not "broadly on track".

- **Time.** Financial close by month 18; mechanical completion by month 44; commercial operation no later than December 2030 (month 48).
- **Cost.** Electrolyser system at or below US$850/kW installed; total capital at or below ₹1,000 crore including financing costs during construction.
- **Performance.** Year-one output of at least 9,000 tonnes; electrolyser energy use no worse than 53.2 kWh/kg at the start of life; plant availability of at least 97%.
- **Finance.** Minimum DSCR of 1.30 or more; loan-life cover of 1.5 or more; equity IRR at or above 14% in the base case.
- **Green.** Twelve-month rolling emissions at or below 2 kg CO_2e per kg, verified by an accredited agency [?].

1.3 Assumptions and constraints

- Exchange rate ₹90 per US dollar for project costs and ₹87 for the 2025 auctions ESTIMATED. A weaker rupee raises the cost of imported stacks; we hedge it (Section ??).
- Delivered renewable power US$32–48/MWh, from a solar-plus-storage price of ₹2.86/kWh [?] to firm, round-the-clock renewable power at ₹5.25/kWh [?] REPORTED.
- Debt at 70% of cost, 9% interest, 15 years; corporate tax 25.17% ESTIMATED.
- **Constraint:** the refinery needs a steady flow every hour; the sun and wind do not provide one. Everything in Section ?? follows from this mismatch.

1.4 Who matters, and how much

A stakeholder is anyone who can change the outcome or is changed by it. We map them on two axes: how much power they have over the project, and how much they care about it. The quadrant tells us how to treat them.

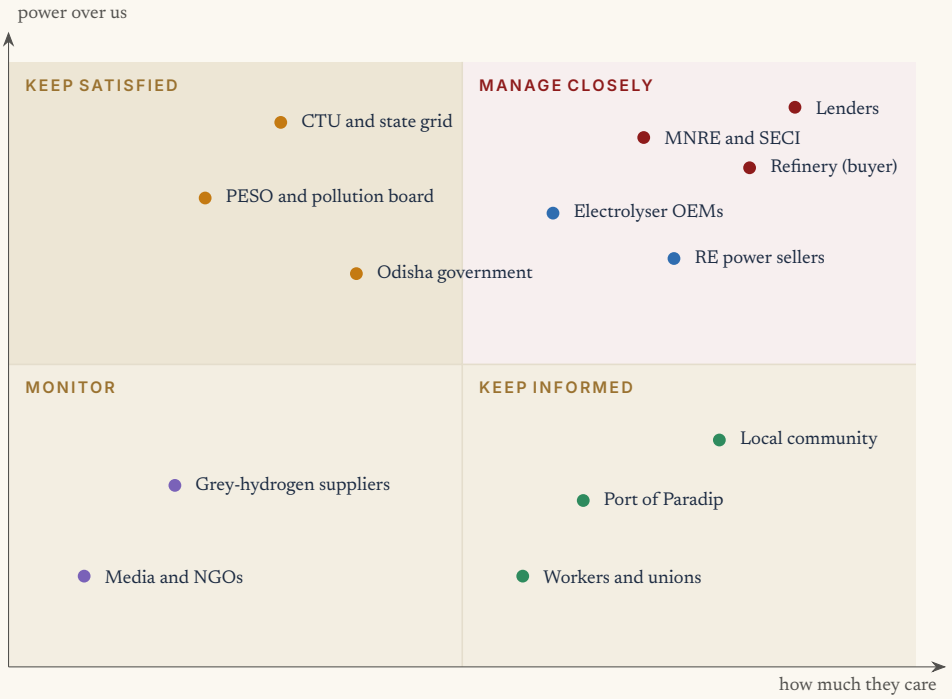


Figure 1. *Power–interest grid. Colour shows the type of stakeholder (red: money and policy; blue: technology and power supply; ochre: permits; green: place and people; violet: outsiders).*

STAKEHOLDER	WHAT THEY WANT	WHAT THEY CAN DO TO US	HOW WE ENGAGE
Lenders	DSCR $\geq$ 1.30, completion guarantees, proven technology, a creditworthy buyer	Refuse to lend, or lend only on harsh terms; trigger a cash lock-up if cover falls	Lender's engineer from day one; open model; monthly reports
Refinery (buyer)	Reliable flow, purity, a price below its carbon-adjusted grey cost, compliance with any green-hydrogen obligation	Walk away before signing; impose penalties for shortfall	Long-term contract with take-or-pay; joint operating committee
MNRE and SECI	Mission targets met; projects that actually get built	Change incentive or waiver rules; delay certification	Early registration; data sharing on cost
Electrolyser OEMs	Orders, references, the right to blame the power supply for any wear	Late delivery; warranties that exclude the real failure modes	Performance guarantees on kWh/kg and degradation; liquidated damages
RE power sellers	A creditworthy buyer for 25 years	Curtail; fail to build on time	PPAs with delivery guarantees; two sellers, not one
CTU and state grid	Grid safety; transmission paid for	Delay the connection, the longest task on our plan	Apply early; pay for studies before financial close

STAKEHOLDER	WHAT THEY WANT	WHAT THEY CAN DO TO US	HOW WE ENGAGE
PESO and pollution board	Safe storage of a flammable gas; water and effluent compliance	Withhold licences	Design to code from FEED; pre-application meetings
Local community	Jobs, water, safety	Protest; slow land access	Water from treated effluent, not the river; local hiring targets

WHAT THE STAKEHOLDER MAP TELLS US

The four stakeholders in the top-right box (lenders, the buyer, the ministry and SECI) all care about one thing more than any other: **that the project is bankable**. That means the finance section is not a side-show to project management; it *is* stakeholder management for the people with the most power. The permit-givers in the top-left box care little but can stop everything, so they get early, boring, thorough paperwork. MEDIUM

Choosing the site with a method, not a hunch

Four coastal hubs, six criteria, and a test of how easily the answer flips.

Location is the one operations decision you cannot undo. We used the **Analytic Hierarchy Process (AHP)**, a standard tool for choosing between sites [?]. It works in three steps. First, we compare the criteria two at a time ("is being near the buyer more important than being near the port, and by how much?") on a 1–9 scale. Second, a little matrix algebra turns those comparisons into weights and checks that our judgements do not contradict each other. Third, we score each site on each criterion and add up the weighted scores.

2.1 The weights

CRITERION	WEIGHT	WHY IT MATTERS FOR *OUR* PRODUCT
Anchor buyers nearby	33.3%	Hydrogen is costly to move. A buyer within a short pipeline run removes trucks, compression and a whole layer of risk.
Renewable resource	19.5%	Power is about half of our cost. Better sun and wind means fewer megawatts to buy and a higher electrolyser load.
Grid and transmission	19.5%	We bring solar and wind power across the state over the national grid; a weak node delays connection.
Port and export	10.7%	Matters for phase two (ammonia) and for importing stacks.
Land and state policy	10.7%	Land cost, speed of clearances, state incentives.
Water	6.4%	About 9 litres of pure water per kg in the reaction alone, more once purification and cooling are counted [?]; small in cost but sensitive locally.

The consistency ratio is **0.006**, far below the 0.10 ceiling Saaty set [?]. In plain words, our pairwise judgements hang together; we did not, for example, say A beats B, B beats C and C beats A. HIGH (for the arithmetic; the judgements themselves are ours, so MEDIUM.)

2.2 The scores and the result

SITE	BUYERS	RE	PORT	WATER	LAND	GRID	THE REASON IN ONE LINE
Paradip	9	5	8	7	7	7	IOCL refinery, IFFCO and Paradeep Phosphates within a few km; Odisha wind is weak.
Kandla	6	9	8	4	8	8	Best solar and wind in India; but IFFCO Kandla buys ammonia, and the refineries sit across the Gulf of Kutch; water-stressed.
Kakinada	6	7	7	7	7	7	Coromandel nearby; a crowded green-ammonia cluster competing for land and power.
Tuticorin	5	8	8	6	7	6	Strong Tamil Nadu wind; few anchor buyers for hydrogen itself.

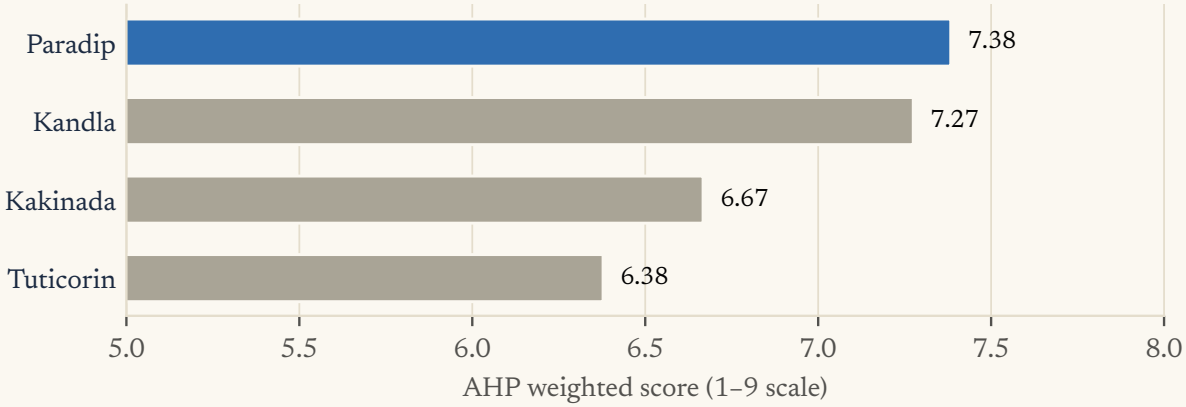


Figure 2. *AHP weighted scores. Paradip wins, but only just. Source: our model,* `siteselect.py`.

2.3 How fragile is the answer?

A ranking is only as good as its closest call. We asked: *how low would the weight on "buyers nearby" have to fall before Kandla wins?* The answer is **30.8%**, against the 33.3% we chose. A shift of 2.5 percentage points flips the decision.

> **WHAT THE SITE ANALYSIS REALLY SAYS**
>
> The site depends on the product. **If we sell hydrogen, Paradip wins, because the buyer is next door. If we sell ammonia or export, Kandla wins, because cheap power then matters more than proximity.** That is why we wrote the product decision first. It also tells us what to watch: if the Paradip refinery will not sign, we should not "make Paradip work"; we should move the project to the best-power site and change the product. MEDIUM

> **A LOCAL WRINKLE**
>
> Paradip is not empty. The lowest green-ammonia bid in India's 2025 auctions, ₹49.75/kg, was for IFFCO at Paradeep [?]. A cluster brings shared pipelines and skilled workers, but also competition for the same land, water and transmission capacity. We treat "cluster congestion" as a named risk in Section ??.

The schedule, and the deadline hiding inside it

A work breakdown, a critical path, 50,000 simulated timelines, and a policy cliff on 31 December 2030.

3.1 Breaking the work down

A work breakdown structure (WBS) splits the project into packages small enough to cost, schedule and give to one owner. Ours has six branches.

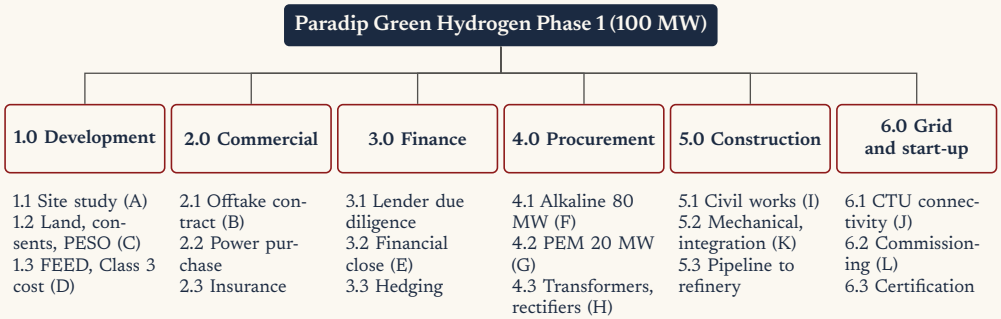


Figure 3. *Work breakdown structure. Letters in brackets link each package to an activity in the schedule network.*

3.2 The network and the critical path

Each activity has three duration estimates: optimistic, most likely and pessimistic. Lead times for transformers and electrolysers follow what Dossier 1 found in the market; the grid connection (J) gets the widest range because it is the task developers complain about most [?]. ESTIMATED

ID	ACTIVITY	MONTHS (O/M/P)	NEEDS	PERT MEAN	SLACK	CRITICAL IN
A	Pre-feasibility and site selection	3 / 4 / 6	–	4.2	0	100%
B	Offtake negotiation, 25-year contract	6 / 9 / 15	A	9.5	0	46%
C	Land, consents, PESO approvals	6 / 9 / 14	A	9.3	0	41%
D	FEED to AACE Class 3 estimate [?]	6 / 8 / 12	A	8.3	1	13%
E	Financing and financial close (FID)	3 / 5 / 9	B, C, D	5.3	0	100%
F	Alkaline electrolysers, 80 MW	10 / 14 / 20	E	14.3	0	45%
G	PEM electrolysers, 20 MW	8 / 11 / 16	E	11.3	3	3%
H	Transformers and rectifiers	10 / 14 / 22	E	14.7	0	**52%**
I	Civil works and site preparation	8 / 10 / 14	E	10.3	4	0.3%
J	RE contracts and CTU connectivity	12 / 18 / 30	B	19.0	9	0.3%
K	Mechanical installation, integration	6 / 8 / 12	F, G, H, I	8.3	0	100%
L	Commissioning, performance tests	3 / 4 / 7	K, J	4.3	0	100%

Using the most-likely durations, the critical path (the longest chain, where any delay delays the whole project) runs **A** → **B/C** → **E** → **F/H** → **K** → **L** and finishes in **44 months**. Several branches tie, which is the first warning sign.

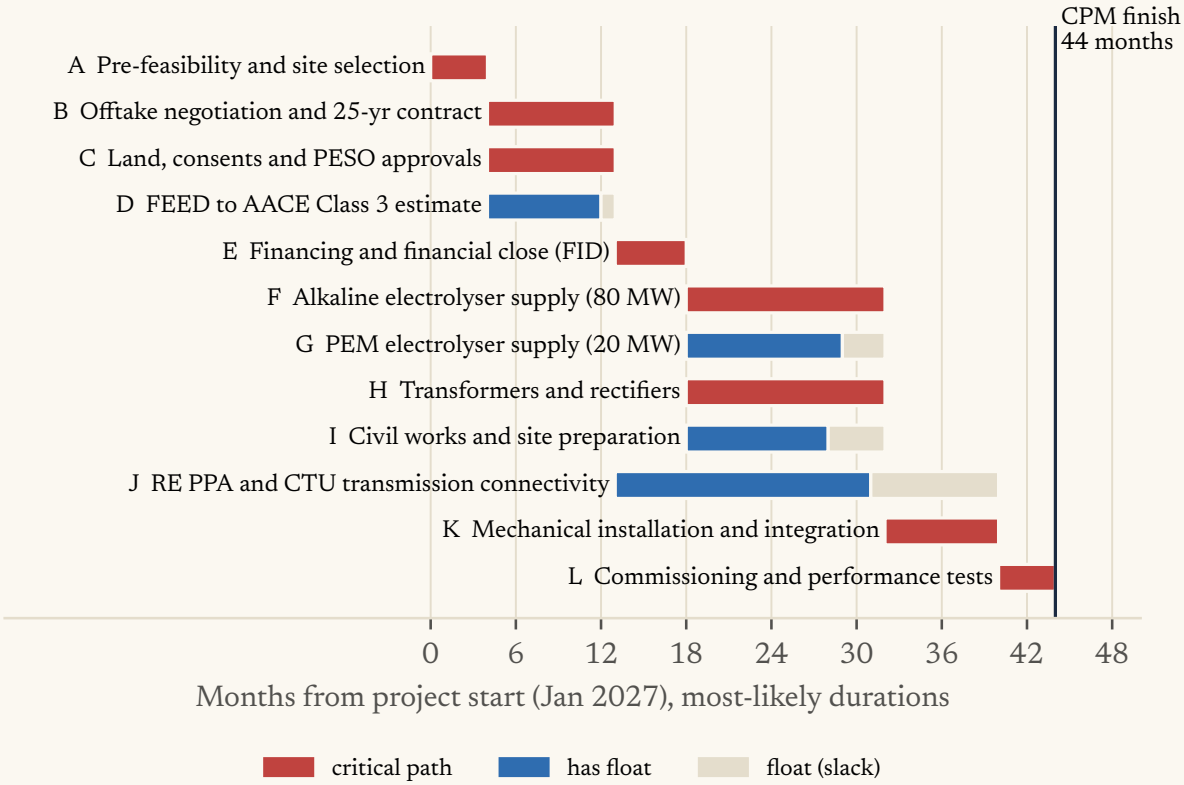


Figure 4. *Gantt chart at most-likely durations. Red bars are critical; grey shows float. Source:* `schedule.py`.

3.3 Why the plan will probably run late: merge bias

The classic PERT method adds up the expected durations along one path [?]. It quietly assumes only one path can be late. In a network where several paths meet at one point (here, at financial close E and at installation K), the project waits for the *slowest* of them, and the slowest of several uncertain things is later, on average, than any one of them. This is called merge bias, and it was spotted only five years after PERT was invented [?].

So we simulated the project 50,000 times, drawing every duration from a Beta-PERT distribution.

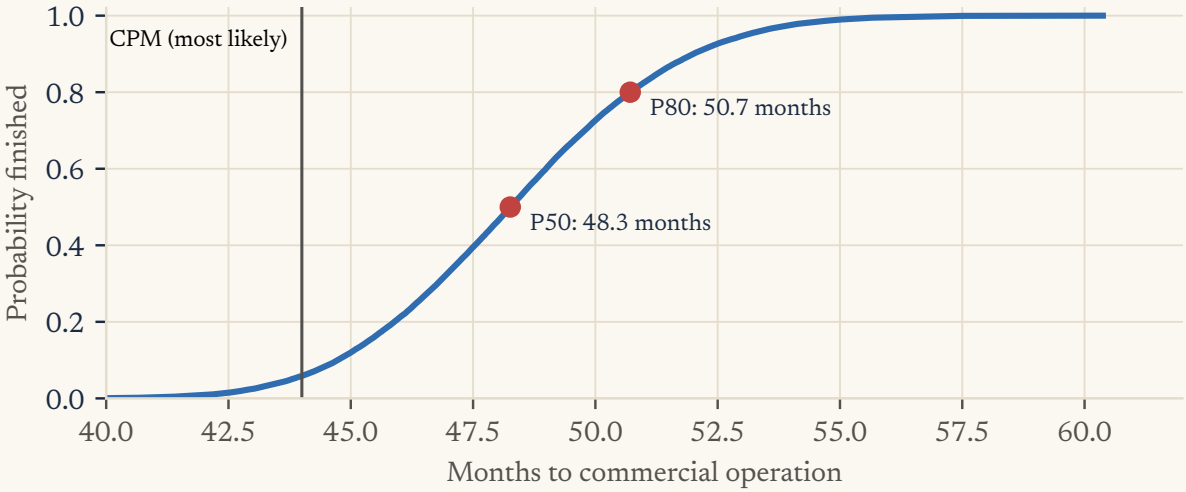


Figure 5. *Distribution of completion times across 50,000 simulated projects. Source:* `schedule.py`.

MEASURE	MONTHS FROM JANUARY 2027
Critical-path plan (most likely durations)	44.0
P50 (half the simulated projects finish by then)	48.3
P80 (the date we would promise a lender)	50.7
P90	52.0
Merge bias (P50 minus plan)	+4.3

THE LONG POLE IS THE TRANSFORMER, NOT THE ELECTROLYSER

Across the simulations, the transformers and rectifiers (H) lie on the critical path 52% of the time; the alkaline stacks (F) 45%. Grid-scale transformers have become a global bottleneck because every renewable and data-centre project wants them at once. A project manager who watches only the electrolyser order is watching the wrong machine. MEDIUM

3.4 The deadline hiding inside the schedule

India waives inter-state transmission charges for 25 years for green hydrogen plants commissioned by **31 December 2030** [?] REPORTED. Starting in January 2027, that date is month 48. Our P50 is month 48.3.

THE SCHEDULE IS A FINANCE PROBLEM

As planned, the plant has only a 46% chance of being commissioned before the transmission waiver closes. Missing it would add a transmission charge to every unit of power we buy for 25 years. If that charge were even ₹0.40/kWh, the price we need would rise by about ₹23/kg (each US$1/MWh of power costs about ₹5.3/kg in our model) ESTIMATED LOW (for the size of the charge).

What we can do about it, tested in the model:

- **Release long-lead orders before financial close.** A "limited notice to proceed" lets the transformer and electrolyser makers start work three months before the loan is signed, with a reservation payment at risk. That lifts the chance of meeting the deadline from **46% to 82%** (two months: 72%; four months: 89%) and pulls the P80 to month 47.8. The deposit, perhaps 10% of the electrolyser order (about ₹75 crore), is lost only if the project dies before close. The 36-point gain in probability, against a waiver worth on the order of ₹200 crore in present value, is worth about ₹70 crore; the bet is favourable as long as the chance of the project dying after the deposit is low. ESTIMATED

- **Run land and offtake in parallel with a time box.** B and C are each critical about 40–46% of the time; neither can be allowed to drift.
- **Put P80, not the plan, in the contracts.** The buyer and lenders get month 51 as the guaranteed date, with liquidated damages from the EPC contractor for every week late beyond that.

> **EARNED VALUE: HOW WE WILL TRACK IT**
>
> Once construction starts we report three numbers each month: what we planned to have done (planned value), what we have actually done (earned value) and what it cost (actual cost). Two ratios fall out. The schedule performance index (earned over planned) tells us whether we are ahead or behind; the cost performance index (earned over actual) tells us whether we are getting a rupee of work for each rupee spent. An index below 0.95 for two months running goes to the steering committee.

Operations: matching a restless supply to a steady customer

Capacity, buffer inventory, maintenance, quality, supply chain, and a forecast of how fast India's buyers will switch.

Every operations problem in this plant comes from one mismatch. The refinery wants roughly the same amount of hydrogen every hour of every day. The sun shines for about a third of the day and the wind blows when it pleases. An operations manager has three ways to bridge a gap like this: hold *capacity* (build more than you need), hold *inventory* (store the product) or buy *flexibility* (pay for a back-up). The textbook calls these the levers of aggregate planning; we use all three and let a model choose the mix.

4.1 The dispatch model

We wrote an hourly linear programme (a model that finds the cheapest plan meeting every constraint) for a full year of 8,760 hours. It chooses how many megawatts of solar and wind to contract, how big the hydrogen store should be, and how much firm back-up power to buy, so that the refinery gets exactly its 9,000 tonnes a year at the lowest total cost. It is solved with the HiGHS solver [?] through SciPy [?]. The solar and wind profiles here are synthetic, calibrated to typical Odisha capacity factors (solar 20.3%, wind 33.7%); the repository contains functions to swap in real hourly data from PVGIS and NASA POWER [?, ?]. So the shape of the answer is solid; the exact megawatts are illustrative. MEDIUM

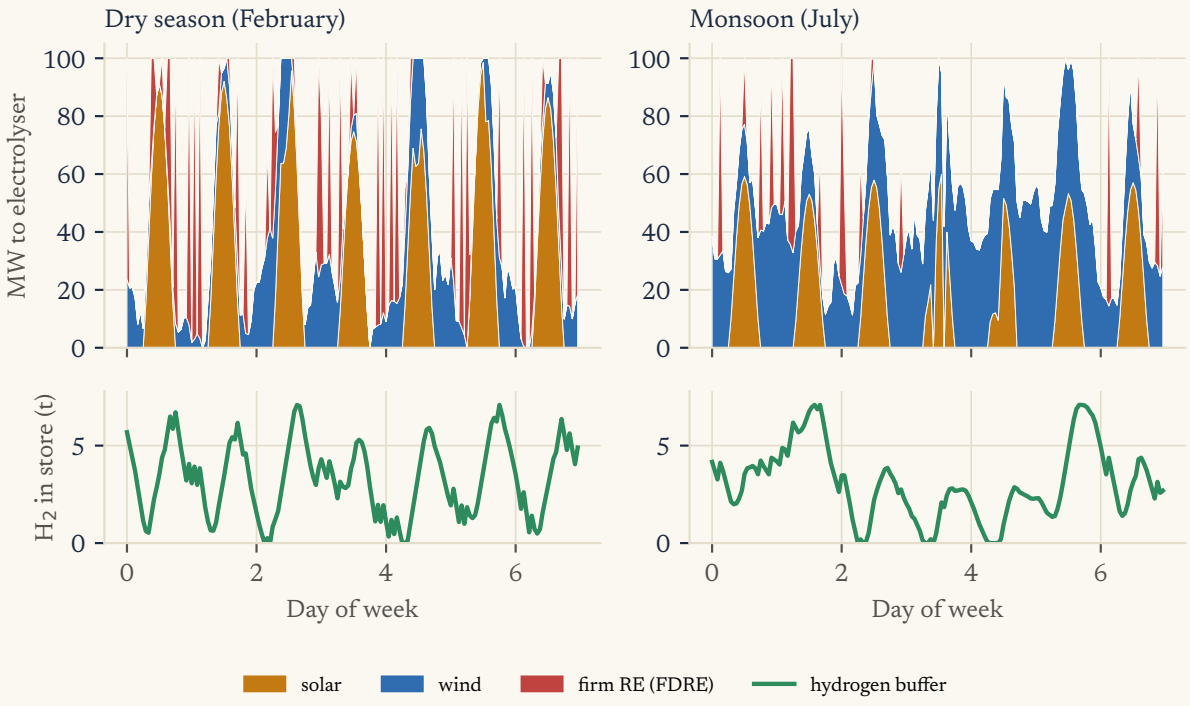


Figure 6. *One week in the dry season and one in the monsoon. Top: what powers the electrolysers each hour. Bottom: tonnes of hydrogen in the buffer. Source:* `dispatch.py`.

OUTPUT	VALUE	WHAT IT MEANS
Solar contracted	104.7 MW	Cheap daytime energy, the bulk of supply.
Wind contracted	67.8 MW	Fills evenings and the monsoon, when solar sags.
Renewable MW per electrolyser MW	1.72	We buy nearly twice the nameplate so the stack runs more hours.
Hydrogen buffer	7.1 t	About 7 hours of refinery demand: safety stock, in tonnes of gas.
Firm back-up power share	21.4%	Bought at ₹5.25/kWh [?]: expensive, but cheaper than idle stacks.
Electrolyser load factor	54.7%	The stacks run about 4,800 full-load hours a year.
Renewable energy curtailed	2.6%	Some surplus is thrown away; storing it all would cost more.
Delivered power cost	US$47.9/MWh	Our finance model's base uses US$38; this is a cautious cross-check.

INVENTORY IS CHEAPER THAN IDLE CAPACITY, UP TO A POINT

The model buys only about 7 tonnes of storage, because compressed hydrogen storage is expensive per tonne. It prefers to oversize the renewables and pay for some firm power. The hydrogen buffer plays the role of a classic decoupling inventory: it lets a variable upstream process feed a level downstream one. The firm power plays the role of an expedited supplier: used sparingly, at a premium. MEDIUM

4.2 The quality constraint nobody mentions: what "green" allows

India's Green Hydrogen Standard caps emissions at 2 kg of CO_2e per kg of hydrogen, averaged over twelve months [?]. India's grid emitted 0.675 kg of CO_2 per kWh in 2025–26 [?] REPORTED. At 53.2 kWh per kg, that means **grid power can supply at most about 5.6% of our electricity** before we lose the right to call the product green, even if everything else were zero-carbon ESTIMATED.

AN OPERATIONS RULE THAT IS REALLY A COMPLIANCE RULE

Our back-up power must be *certified renewable* round-the-clock power, never plain grid power. If the 21% back-up share came from the ordinary grid, our hydrogen would carry about 7.7 kg of CO_2 per kg, nearly four times the limit. Metering, hourly matching and the audit trail are therefore part of the plant's quality system, not paperwork bolted on at the end. HIGH

4.3 Maintenance and wear

Electrolyser stacks get slowly less efficient: we assume energy use creeps up 0.6% a year until the stack is replaced. At our load, the alkaline stacks are replaced in **year 14** and the PEM stacks in **year 11**, each costing about 35% of a new system ESTIMATED. Two decisions follow.

- **A major-maintenance reserve.** Lenders do not let one year's cash flow absorb a ₹200–300 crore swap of the alkaline stacks. We set aside cash every year into a reserve account (MMRA) so the cost is spread evenly over each stack's life. In the model, this alone turned a negative cover ratio in the swap year into a smooth one.
- **Condition-based, not calendar-based, replacement.** We monitor cell voltage stack by stack. A stack is replaced when the cost of the extra electricity it wastes exceeds the cost of carrying the new stack earlier, not when a calendar says so. This is the same logic as an economic replacement policy for any machine.

We track one headline number, borrowed from manufacturing's overall equipment effectiveness: **hydrogen effectiveness = availability × efficiency ratio × on-spec ratio**. Availability is the share of hours the plant could run; the efficiency ratio is design kWh/kg divided by actual kWh/kg; the on-spec ratio is the share of output meeting the purity and green-certification rules. It turns three separate worries into one number on the control-room wall.

4.4 Supply chain: who we cannot afford to depend on

We sort what we buy by how much it costs and how hard it is to replace, the Kraljic matrix of purchasing.

CATEGORY	ITEMS	STRATEGY
Strategic (costly, hard to replace)	Alkaline stacks; transformers and rectifiers	Two qualified suppliers each; performance guarantees on kWh/kg and degradation; spare stack modules held on site.
Bottleneck (cheap, hard to replace)	Iridium-coated PEM components; membranes	Buy-back and recycling clause; a frame contract with the OEM. Iridium is the PEM's weak link [?].
Leverage (costly, easy to replace)	Compressors, water treatment, pipeline	Competitive tender; standard specifications.
Routine	Instruments, consumables	Catalogue buying; minimum stock rules.

4.5 Demand forecasting: how fast will Indian buyers switch?

There is almost no past to fit a time-series model to: India's green hydrogen demand today is a few thousand tonnes. For a new product with no sales history the standard tool is the **Bass diffusion model** [?], which says adoption is driven by two forces: an outside push (here, mandates, auctions and SIGHT incentives) and copying (buyers switching once they see peers do it). We let every parameter vary across a range and ran 20,000 futures. Base volumes come from the last detailed national mapping, which found about 2.6 Mt a year going into refineries out of 5.6 Mt made at home in 2018–19 [?]. ESTIMATED LOW

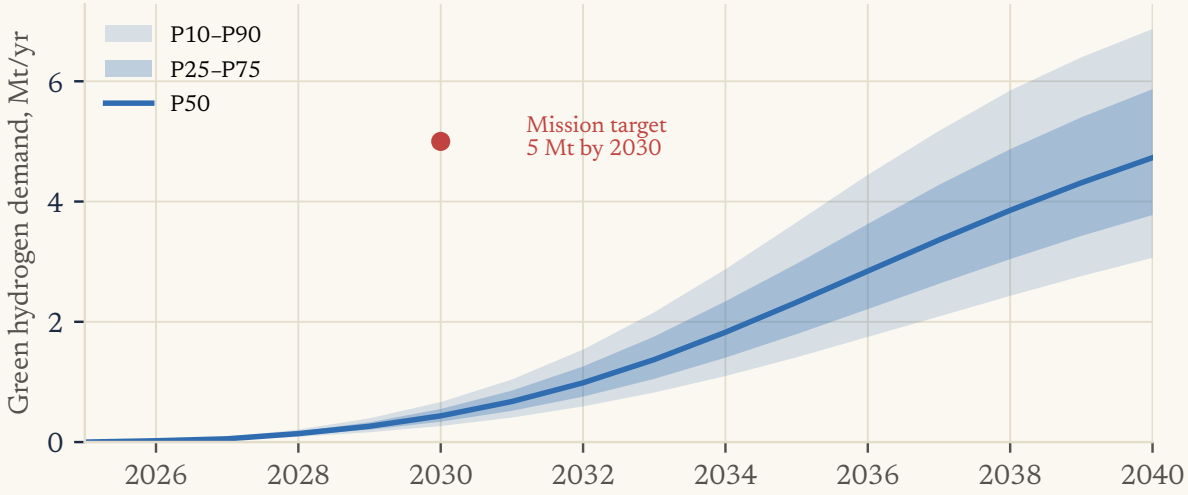


Figure 7. *Forecast of green hydrogen demand in India, all uses, 20,000 Bass-diffusion futures. Source:* `demand.py`.

GREEN HYDROGEN DEMAND, MT PER YEAR	P10	P50	P90
All uses, 2030	0.27	0.44	0.67
All uses, 2035	1.41	2.32	3.66
Refining only, 2030	0.04	0.10	0.22
Refining only, 2035	0.20	0.49	1.02

> **WHAT THE FORECAST IS FOR**
>
> Two lessons, neither of which depends on the exact numbers. First, **not one of the 20,000 futures reaches the Mission's 5 million tonnes by 2030** [?]; on our ranges, that target is a 2040 story. Second, our plant's 9,000 tonnes would be about **9% of the likely green refinery market in 2030**. In a market that thin, a plant without a signed buyer is a plant without a market. Build only behind a contract. LOW (for the levels), MEDIUM (for the lessons).

Once the plant runs, forecasting becomes a different job: predicting next week's hourly power and the refinery's daily draw to plan the buffer. That calls for seasonal time-series models fed by weather forecasts; the repository's data plan (Section **??**) lists the feeds.

Finance: the plant as a lender sees it

How the deal is built, why the first design fails, the five moves that rescue it, and the odds that the rescue holds.

This is the section where we stop being project managers and become what we really are: two finance students asking whether the cash will turn up. We built the plant as a lender would, in a special-purpose company with its own cash-flow waterfall, and then ran it 20,000 times.

5.1 How the deal is built

Project finance lends against the project's own cash flows, not against the sponsor's balance sheet [?]. That makes every contract around the SPV a piece of risk being handed to whoever can carry it best.

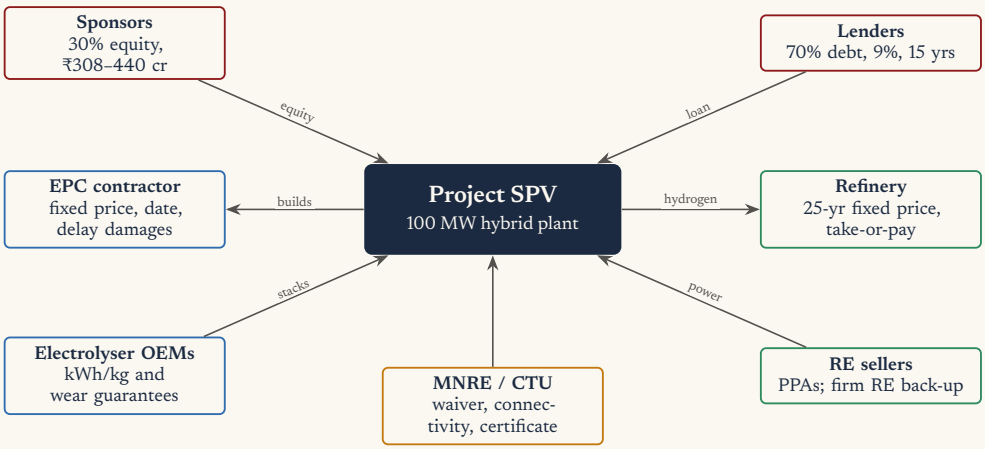


Figure 8. *The contract map. Each arrow is a contract that moves a risk off the SPV: construction risk to the EPC contractor, performance risk to the OEMs, price and volume risk to the refinery, resource risk to the power sellers.*

The cash waterfall each year is simple to say. Revenue, minus power, operations, insurance and water, gives **EBITDA**. Minus tax and the maintenance reserve gives **cash flow available for debt service (CFADS)**. Divide CFADS by that year's interest plus principal and you have the **debt service cover ratio (DSCR)**, the number every lender watches. Lenders want a minimum near 1.30: thirty paise of spare cash for every rupee owed.

5.2 What a kilogram costs, by stack

Our levelised-cost model (Dossier 1's engine, updated) puts the hybrid at US$4.02/kg in the base case and US$4.25 at the median of 100,000 simulations. That is about US$0.19 above a pure alkaline plant: the price of fast ramping and of the AEM option bay. RMI's 2025 India study puts stand-alone on-site projects at US$4.4–4.8/kg [?], close to our range. Peer-reviewed work agrees that the electricity price and the load factor drive most of the uncertainty [?, ?]. MEDIUM

STACK, US$/KG OF HYDROGEN	BASE	P10	P50	P90
Alkaline (pressurised)	3.83	3.63	4.07	4.58
Hybrid 80/20 (our choice)	**4.02**	**3.79**	**4.25**	**4.76**
PEM	5.09	4.71	5.32	5.99
AEM	5.83	5.22	6.43	8.01
SOEC	8.93	8.02	9.55	11.60

5.3 The first design fails

At the IOCL price of ₹397/kg, the plant as first costed (US$1,250/kW installed, power at US$38/MWh) does not work.

MEASURE	BASE, ₹397	BASE, ₹279	TARGET, ₹397	TARGET, ₹279
Capital cost incl. interest during build (₹ cr)	1,389	1,389	973	973
Debt / equity (₹ cr)	1,028 / 440	1,028 / 440	718 / 308	718 / 308
Hydrogen, year 1 (t)	9,655	9,655	9,655	9,655
Revenue, year 1 (₹ cr)	383	269	383	269
Power bill, year 1 (₹ cr)	182	182	153	153
EBITDA, year 1 (₹ cr)	163	49	202	88
Project IRR	4.3%	n/a	**12.2%**	1.0%
Equity IRR	−0.3%	n/a	**18.6%**	−6.0%
Minimum DSCR	0.63	−0.12	**1.36**	0.45
Loan-life cover ratio	0.87	0.04	1.63	0.65
NPV at 10% (₹ cr)	−515	−1,377	+166	−531

The power bill alone eats almost half of revenue. The base design cannot repay its loan in any year: a DSCR of 0.63 means the plant earns 63 paise for every rupee of debt service due. To clear a 1.30 cover and a 14% equity return, it would need **₹522/kg**, about a third more than any Indian buyer has paid. HIGH (that the base design fails), MEDIUM (for the exact tariff).

5.4 What you have to believe

So we ran the problem backwards. Instead of asking “what does the plant earn?”, we asked “what price does the plant need, and which moves bring that price down?” Each bar in the ladder below adds one design-to-cost move to the ones above it.

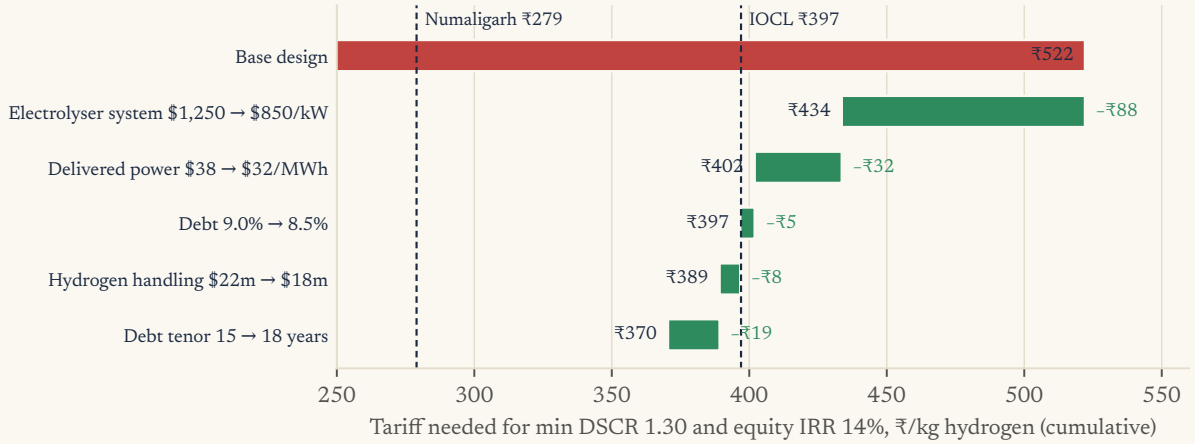


Figure 9. *The lever ladder: required tariff after each cumulative move. Source:* `finance.lever_ladder()`.

MOVE	SAVES (₹/KG)	NEEDED AFTER	IS IT BELIEVABLE?
Electrolyser system US$1,250 → 850/kW	88	434	Yes, if we buy competitively: Chinese alkaline system prices roughly halved in three years [?]. The biggest lever by far. MEDIUM
Delivered power US$38 → 32/MWh	32	402	Plausible with an oversized solar-wind hybrid and storage near ₹2.86/kWh [?]. MEDIUM
Debt 9.0% → 8.5%	5	397	Needs a strong sponsor or a green-loan programme. MEDIUM
Handling US$22m → 18m	8	389	A shorter pipeline and a smaller buffer. MEDIUM
Loan tenor 15 → 18 years	19	370	Common for contracted infrastructure if the buyer is strong. MEDIUM

THE TWO NUMBERS THAT DECIDE EVERYTHING

The stack price and the power price account for ₹120 of the ₹152/kg gap. Financing tweaks matter, but they cannot rescue a plant that pays too much for its machines and its electrons. This is the most useful sentence in the book for a procurement team. And note what the ladder cannot reach: Numaligarh's ₹279/kg [?]. Even with every move, it sits ₹91 below what our plant needs. Either that bidder knows something we do not (captive power, very cheap Chinese stacks, a parent willing to accept a low return), or it has under-priced. Section **??** asks the same question of the ammonia auctions. MEDIUM

5.5 The cover ratio, year by year

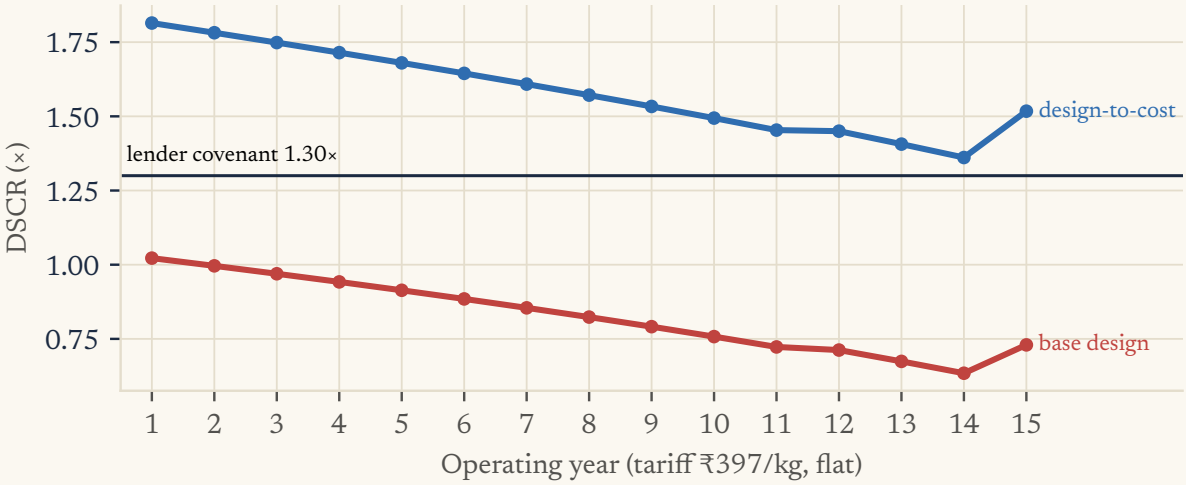


Figure 10. *DSCR by year at ₹397/kg, base design versus design-to-cost target, 15-year annuity loan. Source:* `finance.py`.

The target design starts at 1.81 and slides to 1.36 by year 14. The slide has two causes: stacks wear (more power per kilogram) while the price stays fixed, and operating costs rise 4% a year while revenue does not. A flat 25-year price with rising costs is a slow squeeze. The jump in year 15 is the alkaline stack swap being absorbed by the reserve built up before it.

5.6 Twenty thousand futures

A single base case hides the shape of the risk. We drew every uncertain input from a range (capital cost, power price, load hours, stack efficiency, stack life, interest rate) and added a construction overrun that is skewed upwards, because overruns in large projects are far more common than underruns [?]. Our overrun has a median of +5% and a P80 of +16%.

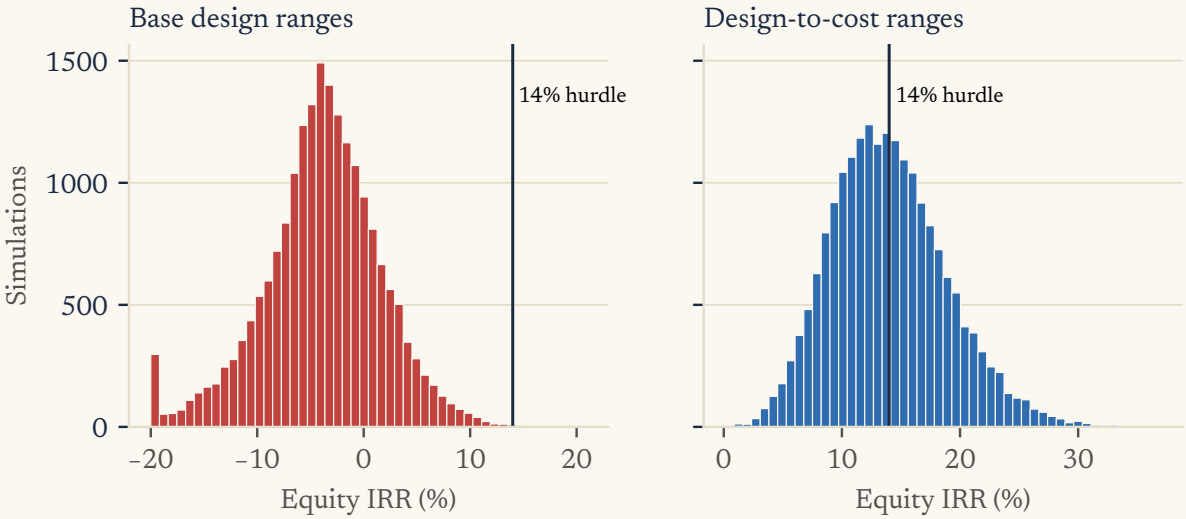


Figure 11. *Equity IRR across 20,000 simulated futures at ₹397/kg. Left: base design ranges. Right: design-to-cost ranges. Source:* `finance.monte_carlo`.

AT ₹397/KG	BASE DESIGN	TARGET, 15-YR LOAN	TARGET, 18-YR LOAN
Equity IRR, P10 / P50 / P90	−10.6 / −3.7 / 2.9%	8.1 / 13.6 / 20.4%	– / 14.7% / –
Project IRR, P50	2.5%	10.4%	–
NPV at 10%, P50 (₹ cr)	−717	+30	–
Minimum DSCR, P50	0.53	1.17	1.29
Chance min DSCR < 1.20 (lock-up)	99.9%	**56.9%**	**32.0%**
Chance min DSCR < 1.00 (default risk)	99.3%	**16.3%**	**5.4%**
Chance equity IRR < 12%	99.8%	36.8%	30.2%

BANKABLE ON PAPER, FRAGILE IN PRACTICE

The design-to-cost plant earns 18.6% for its owners in the base case. But the median of 20,000 futures is 13.6%, and in more than half of them the cover ratio dips below 1.20 at least once, the level at which lenders typically block dividends. **The base case is the optimistic case**, because the things that go wrong (overruns, weak wind, early stack wear) are bigger than the things that go right. Stretching the loan to 18 years is the single cheapest fix we found: it cuts the chance of a covenant breach from 57% to 32% and of a sub-1.0 year from 16% to 5%. MEDIUM

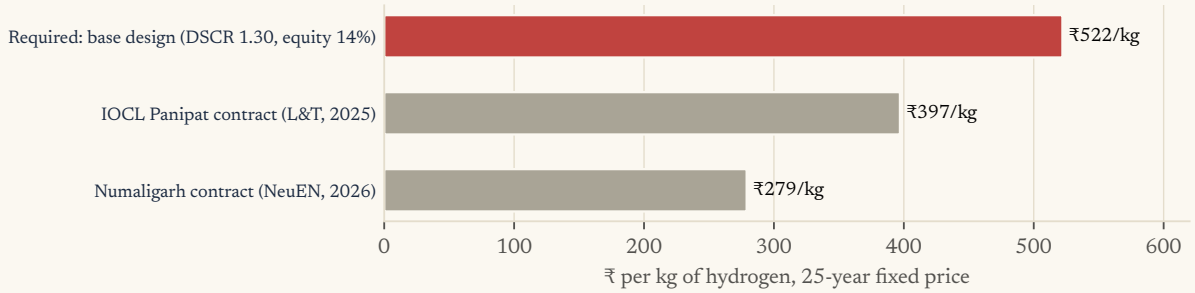


Figure 12. *What the base design needs against what Indian refineries pay. Sources: [?, ?]; our model.*

5.7 What we would ask for at the negotiating table

Every recommendation below moves a specific risk to someone better able to carry it, and each is tied to a number above.

1. **An 18-year loan, sculpted to cash flow.** Repayments shaped to fall when stack wear bites, not a flat annuity. (Covenant breach risk 57% → 32%.)
2. **Partial indexation of the hydrogen price.** Even 1–2% a year on the operating-cost share of the price would offset the 4% cost escalation that drives the DSCR slide.
3. **Performance guarantees with teeth.** The OEMs guarantee kWh/kg and degradation rate, with damages that cover the extra power. Efficiency is about half our cost; an unguaranteed efficiency is an unhedged power price.
4. **A fixed-price, date-certain EPC contract** with delay damages at least equal to a month of debt service, and an owner's contingency at P80 (+16%), not P50.
5. **Currency and rate hedges.** Imported stacks are priced in dollars or euros; forward cover from order to delivery. An interest-rate swap for the floating part of the debt.
6. **Reserve accounts.** Six months of debt service in a reserve account (DSRA), plus the major-maintenance reserve already in the model.

Risk: priced in rupees, not adjectives

A register of fourteen risks, each tied to a number in our models, a heat map, one bow-tie in detail, and how much contingency to hold.

A risk register usually fails in one of two ways: it lists everything with equal weight, or it uses words like "significant" that nobody can act on. We tried to avoid both. Each risk below is scored for likelihood (L) and impact (I) on our 1–5 scale from Dossier 1, and wherever possible the impact is linked to a model result. The owner is the party best able to control it, which is often not us.

ID	RISK	L	I	EVIDENCE AND RESPONSE	OWNER
R1	Capital cost above US$850/kW	4	5	Every US$100/kW adds about ₹22/kg to the price we need (lever ladder). *Mitigate:* competitive tender incl. Chinese and Indian PLI suppliers; Class 3 estimate before FID.	Sponsor
R2	Delivered power above US$32/MWh	3	5	About ₹5.3/kg per US$1/MWh. *Transfer:* fixed-price PPAs; two sellers.	RE sellers
R3	Commissioning after 31 Dec 2030 (waiver lost)	3	4	Only 46% on time as planned; 82% with early release of orders. *Mitigate:* limited notice to proceed.	SPV / EPC
R4	DSCR covenant breach	3	5	57% chance below 1.20 at least once (target design). *Mitigate:* 18-yr sculpted loan, DSRA, indexation.	SPV / lenders
R5	Faster stack wear on variable power	3	4	Swap years 14 (alkaline), 11 (PEM) are assumptions. *Transfer:* degradation guarantee; *mitigate:* PEM takes the fast swings.	OEMs
R6	Buyer does not sign, or signs late	3	5	Our plant is ~9% of the likely 2030 green refinery market. *Avoid:* no FID without signed take-or-pay.	Sponsor
R7	Buyer credit or volume default	2	5	PSU refinery, but 25 years is long. *Mitigate:* payment security (letter of credit), termination payments.	Refinery
R8	Transformer and rectifier delay	4	3	Critical 52% of the time. *Mitigate:* order before FID; two vendors.	EPC
R9	Grid (CTU) connection delay	3	3	9 months of float today; widest range on the plan. *Mitigate:* apply at month 4, not month 13.	SPV
R10	Losing "green" status	2	5	Grid power above 5.6% of supply breaks the 2 kg limit. *Avoid:* certified RTC renewable back-up only; hourly metering.	SPV
R11	Rupee depreciation during procurement	3	3	Stacks priced in foreign currency. *Transfer:* forward cover.	Treasury
R12	Hydrogen safety incident	1	5	Flammable gas next to a refinery. *Mitigate:* PESO-compliant design, HAZOP, separation distances; insurance.	EPC / operator
R13	Cluster congestion at Paradip	3	2	Competing for land, water and transmission with ammonia projects. *Mitigate:* early land lock; shared pipeline corridor.	Sponsor
R14	Policy change (incentives, obligations)	2	3	Waiver rules or a consumption obligation could shift. *Accept* and monitor; contracts with change-in-law clauses.	Government

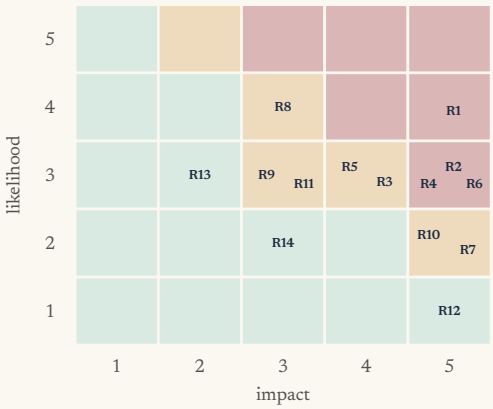


Figure 13. *Heat map. Red cells (score 15+) need a named owner and a funded response before financial close.*

> **WHERE THE RISK REALLY SITS**
>
> The top-right of the map is all money: capital cost (R1), power price (R2), covenants (R4) and the buyer (R6). The technical risks people usually lead with (stack wear, safety) matter, but they are *transferable*, through guarantees and insurance. The financial ones mostly are not. That is why this project needs a risk manager more than it needs another engineer. MEDIUM

6.1 One risk, opened up: the covenant breach

A bow-tie diagram puts one risk event in the middle, its causes on the left and its consequences on the right, with the barriers that stop each one. We use it for R4, because it is where most of the other risks end up.

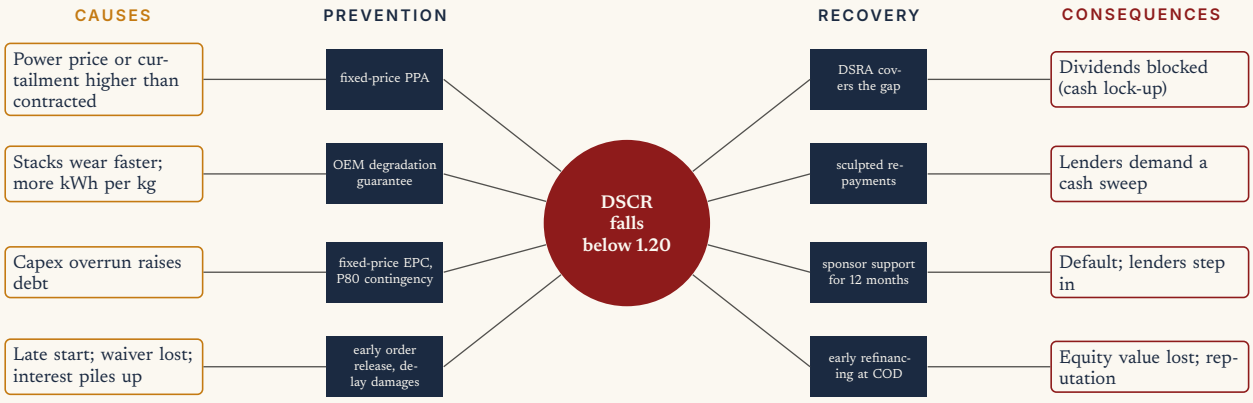


Figure 14. *Bow-tie for the covenant breach. Dark boxes are barriers; each is a contract or an account we can name.*

6.2 How much contingency to hold

Contingency is not padding; it is the gap between the plan and the level of confidence we want. Our simulations give it directly.

CONTINGENCY	P50	P80	WHAT WE HOLD
Schedule (months beyond the 44-month plan)	+4.3	+6.7	7 months in contracts; 3 months bought back by early order release.
Cost (construction overrun)	+5%	+16%	P80 owner's contingency, about ₹155 crore on the ₹973 crore target design.
Revenue (equity IRR shortfall vs. base)	−5.0 pts	−10.5 pts at P10	Held as headroom in the price negotiation, not as cash.

Real options: phase two and the AEM bay

Why the right to expand is worth more than expanding now, what a policy deadline does to that right, and when to switch to AEM.

Net present value asks one question: if we commit today, is it worth it? It cannot value the freedom to *wait and decide later*. In a market where prices, technology and policy are all moving, that freedom is often worth more than the commitment. Real-options analysis prices it, borrowing the logic of financial options [?]. The green hydrogen literature has started to do this seriously: sequential options for power-to-hydrogen plants in Germany and California [?], staged electrolyser investment under technology learning [?], and hydrogen storage as flexibility for wind parks [?].

7.1 Option one: phase two, under a policy cliff

Phase two would add another 100 MW. We model its value as uncertain (30% volatility a year), its cost falling 4% a year as electrolysers get cheaper, and a right to build in any year from 2028 to 2036 with a two-year build. The twist is Indian: a plant commissioned after 31 December 2030 loses the 25-year transmission-charge waiver [?]. We assume that wipes out 12% of phase two's value ESTIMATED LOW. Because the right can be used in any year, it is an American-style option, so we value it with the least-squares Monte Carlo method of Longstaff and Schwartz [?].

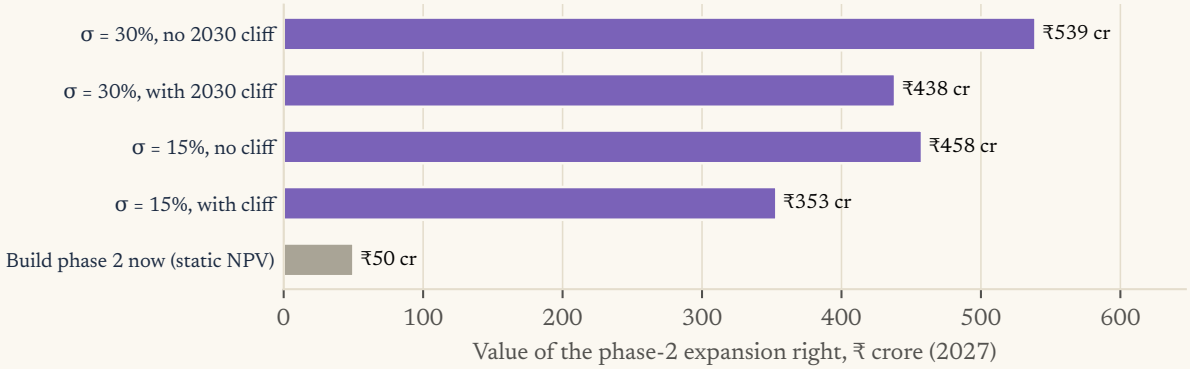


Figure 15. *Value of the right to build phase two, under two volatility assumptions, with and without the 2030 cliff. Source:* `realoptions.expansion_lsmc`.

₹ CRORE, 2027 MONEY	VOLATILITY 30%	VOLATILITY 15%
Build phase two now (static NPV)	50	50
Option value with the 2030 cliff	438	353
Option value without the cliff	539	458
Cost of the cliff	101	105
Never exercised (share of futures)	43%	16%

THREE THINGS THE OPTION MODEL SAYS THAT NPV CANNOT

First, wait. Building phase two today is worth ₹50 crore; keeping the right to build it is worth ₹438 crore. The difference is the value of learning before committing. **Second, the cliff does not make us rush.** In most futures where phase two gets built, it is built late (2036), after the waiver has gone, because rushing in before costs fall is worse than losing the waiver. **Third, the cliff still has a price: about ₹100 crore.** That is the most we should ever pay to make an early phase two possible, for example by buying extra land, oversizing the grid connection, or pre-permitting a second train before 2028. LOW (for the levels), MEDIUM (for the logic).

7.2 Option two: the AEM switch

The 20 MW PEM slot will need new stacks around 2041. At that moment we can buy PEM again, or AEM if AEM has become cheap and, above all, durable. AEM promises PEM-like flexibility without iridium or platinum, but today's evidence on long-term durability is thin, and the leading listed AEM maker is still losing money [?, ?].

We model AEM's cost falling along Wright's law, a fixed percentage for every doubling of cumulative production [?], with a learning rate between 8% and 20% and between three and eight doublings by 2041. Learning in energy models is itself uncertain and should be treated as such [?]. The chance that AEM proves durable (40,000+ hours) rises with deployment. "Keeping the bay AEM-ready" (compatible footprint, power electronics and water quality) costs US$0.3–0.6 million now ESTIMATED.

RESULT	VALUE
Expected value of the switch right (2030 money)	US$1.46 million
Cost of keeping the bay ready	US$0.3–0.6 million
AEM installed cost by the swap year, median	US$777/kW
Chance we switch	59%
Chance AEM proves durable	59%

BUY THE OPTION; LET DURABILITY DECIDE

The option is worth roughly three to five times what it costs to keep, so we buy it. Notice that the chance of switching equals the chance AEM proves durable: in every future where AEM survives in the field, it is also cheaper by 2041. **The gate is not price; it is proof.** So the stage-gate for the switch is written in hours, not rupees: at least 40,000 hours of field data across several sites, an OEM warranty that covers degradation, and a pilot of 1–2 MW in our own bay before we commit the rest. MEDIUM

7.3 The stage-gates

GATE	QUESTION	PASS IF
G0, 2027	Start development?	Buyer term sheet signed; site AHP confirmed with real data.
G1, 2028	Financial close?	Signed take-or-pay; EPC at or below US$850/kW; P80 commissioning before 2031; base-case min DSCR $\geq$ 1.30.
G2, 2028	Protect phase two?	Extra land and grid capacity cost less than ~₹100 crore.
G3, 2033+	Build phase two?	Option model says exercise (value minus cost beats waiting); second buyer signed.
G4, 2039	AEM in the bay?	40,000 h field data; warranty; 1–2 MW pilot passes.

Do India's green-ammonia auctions clear below cost?

The research design, the first results on all 65 bids, the literature it builds on, and what still has to be done before submission.

8.1 The question

In 2025, SECI ran thirteen e-reverse auctions under SIGHT Mode 2A to buy 724,000 tonnes a year of green ammonia for Indian fertiliser plants, on ten-year contracts [?]. The volume-weighted winning price was **₹53.27/kg**. Dossier 1 found that the winning bids quietly assume a hydrogen cost of about US$2.5–2.9/kg, well below what a typical developer can deliver today. That suggests a sharp, testable question:

> ***Do India's green-ammonia auctions clear below the probabilistic cost of delivery? A stochastic levelised-cost and real-options assessment of winner's-curse risk in SECI's SIGHT Mode 2A.***

The *winner's curse* is the old observation that in a competitive auction for something of uncertain cost or value, the winner tends to be the bidder who was most optimistic, and so is systematically likely to have got it wrong [?, ?]. If it is happening here, the policy consequence is serious: contracts that are awarded but never delivered, as has happened to under-priced renewable auctions before. We are not aware of a published study that tests these auctions for it. MEDIUM

8.2 Three hypotheses, three tests

	HYPOTHESIS	TEST
H1	Bids are bunched more tightly than cost differences between bidders would justify.	**Bid structure:** gap between winner and runner-up; dispersion of serious bids.
H2	Winning bids sit below the likely cost of a typical developer.	**Cost coverage:** the probability that each winning bid covers a Monte Carlo cost of green ammonia, for a market-average and for a best-in-class developer.
H3	Some winning bids imply returns below a normal hurdle rate even for the best developers.	**Implied return:** the discount rate at which each bid exactly covers a best-in-class cost (solved numerically).

The cost model is the levelised-cost engine of Dossier 1, extended to ammonia (0.178 t of hydrogen per tonne of ammonia plus a synthesis-loop cost) and net of the SIGHT incentive of ₹8.82, ₹7.06 and ₹5.30 per kg in years one to three [?], which is worth about **₹2.90/kg** levelised over ten years.

8.3 First results

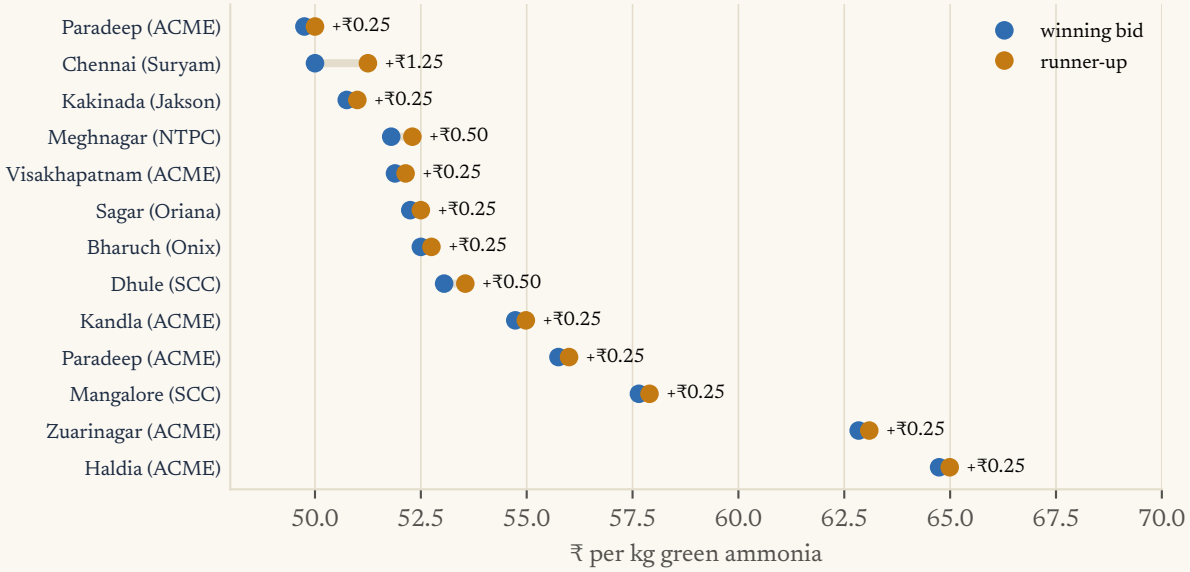


Figure 16. *Winning bid and runner-up in each of the 13 auctions. Data: [?];* `data/raw/seci_mode2a_tranche1_bids.csv`.

H1, bid structure. In ten of thirteen auctions, the runner-up was exactly ₹0.25 behind the winner; the median gap is ₹0.25. ACME won 51% of the volume. The money "left on the table" between winner and runner-up totals only about ₹22 crore a year. *But we must be careful*: in an e-reverse auction, a uniform ₹0.25 gap is most likely the minimum bid decrement, so it tells us bidding was fierce at the end, not that costs were similar. We will therefore lean on H2 and H3, and test H1 against the auction rules once we have the bid logs. MEDIUM

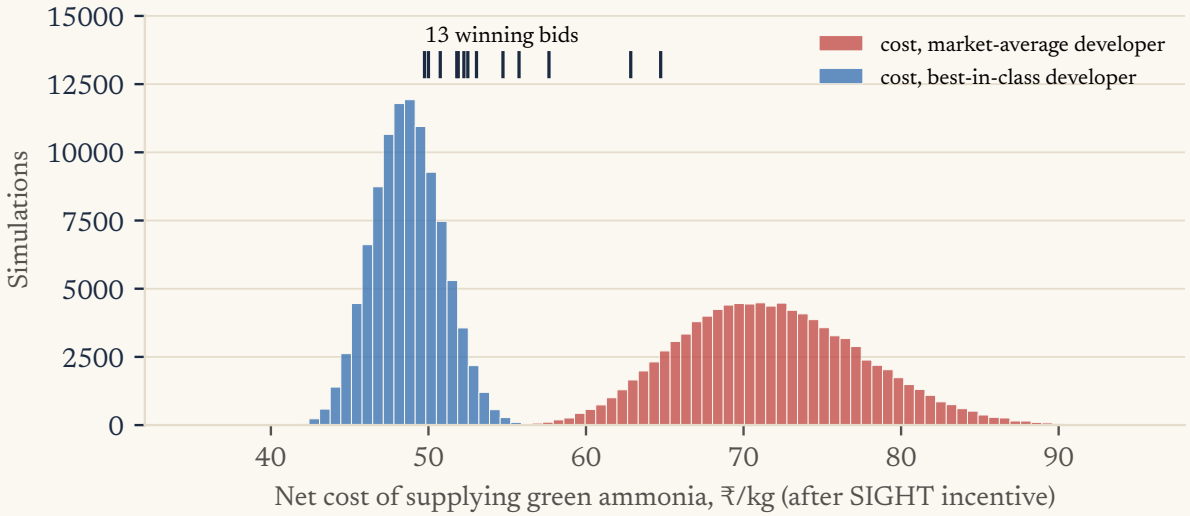


Figure 17. *Simulated net cost of green ammonia (100,000 draws each) against the 13 winning bids. Source:* `auctions.py`.

H2, cost coverage. For a market-average alkaline developer, the median net cost is **₹71.5/kg**; the chance that a winning bid covers it is **0.5%** (volume-weighted). For a best-in-class developer (cheap stacks, cheap power, long-life stacks, low-cost capital), the median falls to **₹48.6/kg** and the coverage rises to **91%**. The auctions are not impossible; they are *only* possible for the best. MEDIUM

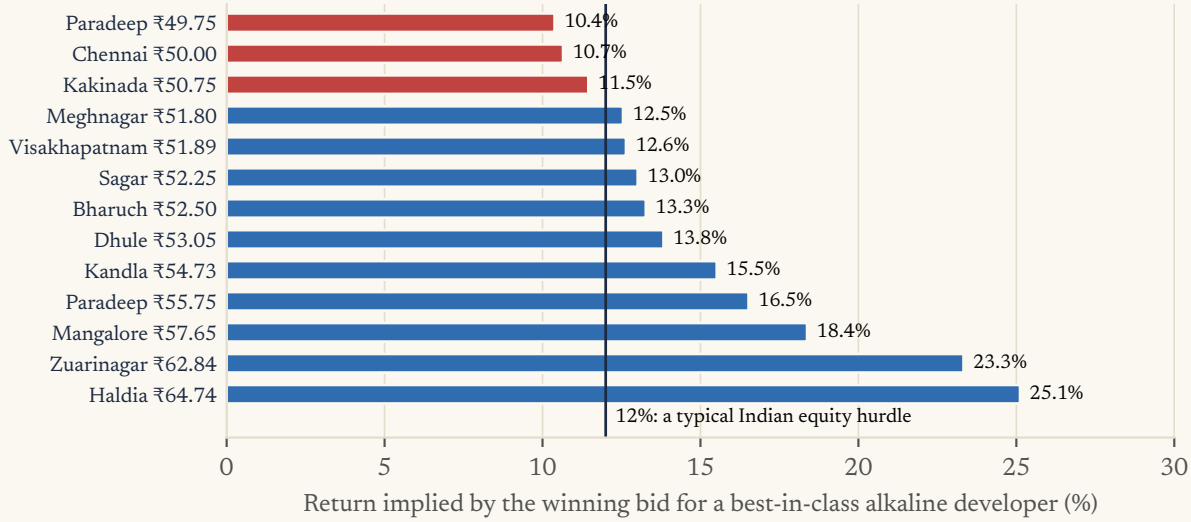


Figure 18. *Return implied by each winning bid for a best-in-class developer. Source:* `auctions.implied_wacc`.

H3, implied return. Even for a best-in-class developer, the implied return ranges from **10.4%** (IFFCO Paradeep, ₹49.75) to **25.1%** (Haldia, ₹64.74), with a volume-weighted average of 14.0%. Three awards (Paradeep IFFCO, Chennai and Kakinada) sit below a typical 12% equity hurdle. Those are where delivery risk is concentrated. MEDIUM

NO.	BUYER, LOCATION (WINNER)	T/YR	WIN ₹	GAP ₹	COVERS: AVG	BEST	IMPLIED
I	IFFCO, Kandla (ACME)	100,000	54.73	0.25	0.0%	99.7%	15.5%
II	IFFCO, Paradeep (ACME)	100,000	49.75	0.25	0.0%	68.9%	**10.4%**
III	Madras Fertilizers, Chennai (Suryam)	4,000	50.00	1.25	0.0%	72.5%	**10.7%**
IV	GNFC, Bharuch (Onix)	50,000	52.50	0.25	0.0%	95.5%	13.3%
V	Paradeep Phosphates, Paradeep (ACME)	75,000	55.75	0.25	0.0%	100%	16.5%
VI	Paradeep Phosphates, Zuarinagar (ACME)	25,000	62.84	0.25	5.2%	100%	23.3%
VII	Indorama, Haldia (ACME)	20,000	64.74	0.25	11.0%	100%	25.1%
VIII	MCF, Mangalore (SCC)	15,000	57.65	0.25	0.2%	100%	18.4%
IX	Coromandel, Visakhapatnam (ACME)	50,000	51.89	0.25	0.0%	92.3%	12.6%
X	Coromandel, Kakinada (Jakson)	85,000	50.75	0.25	0.0%	82.3%	**11.5%**
XI	Krishna Phoschem, Meghnagar (NTPC REL)	70,000	51.80	0.50	0.0%	91.7%	12.5%
XII	Madhya Bharat Agro, Sagar (Oriana)	60,000	52.25	0.25	0.0%	94.4%	13.0%
XIII	Madhya Bharat Agro, Dhule (SCC)	70,000	53.05	0.50	0.0%	97.4%	13.8%

> **WHAT THE FIRST RESULTS MEAN**
>
> India's ammonia auctions have created a market in which **only a best-in-class developer can win and still earn its cost of capital, and a few awards look thin even for them.** That is not proof of a winner's curse; it is the condition under which one would appear. The test that would settle it is time: which awards reach financial close and commissioning on schedule. We will track all thirteen. MEDIUM

8.4 Where the paper sits in the literature

We checked every study in the Gemini literature review before relying on it [?]. Five survive as our core; two were mis-cited; several came from venues we will not cite.

STUDY	WHAT IT DOES	HOW WE USE IT, OR WHAT WE CORRECTED
Glenk & Reichelstein (2019) [?]	Shows when renewable hydrogen reaches cost parity, and how flexibility to sell power raises value.	Missing from the review; our anchor for LCOH logic.
Benalcazar & Komorowska (2022) [?]	Monte Carlo LCOH for Poland; power price and load factor dominate.	Validates our Monte Carlo approach.
Srettiwat et al. (2023) [?]	Techno-economics of solar hydrogen for Belgium.	The review credited this to "Al-Ghussain"; the authors are Srettiwat, Safari, Olcay and Malina (UHasselt).
Fabianek, Glensk & Madlener (2024) [?]	Sequential real options for power-to-hydrogen, Germany and California.	Template for our expansion option.
Trivella (2025) [?]	Electrolyser investment under market and technology uncertainty.	Supports staged investment and the AEM switch.
Kroniger & Madlener (2014) [?]	Hydrogen storage as a real option for wind parks.	Early proof that flexibility has option value.
Zeyen, Victoria & Brown (2023) [?]	Endogenous learning for electrolysers in a European model.	The review mis-attributed this; used for learning-rate ranges.
RMI (2025) [?]	India production pathways: stand-alone on-site at US$4.4–4.8/kg.	The review dated it 2024; it was published in July 2025.
Capen et al. (1971); Thaler (1988) [?, ?]	The winner's curse in high-risk bidding.	Theoretical core of the paper.

The review also proposed a "mean-reverting geometric Brownian motion" for prices. Geometric Brownian motion does not mean-revert; the correct tool is the Schwartz one-factor model, a mean-reverting process on log prices [?], which our `prices.py` implements and tests.

8.5 What remains before submission

- **Data.** The bid logs (if SECI releases them) to test H1 properly; real hourly solar and wind for each auction site; grey-ammonia import prices to fit the price model.
- **Robustness.** Re-run H2 and H3 with each site's own resource profile, with Chinese stack prices, and with the developer's possible captive power.
- **Extension.** Value each award's embedded option to walk away (the bid bond is the strike) to estimate the probability of non-delivery.
- **Target journals.** *Energy Policy* first (the policy question is central); *Energy Economics* if the auction-theory part grows.

The GitHub repository: every number, re-runnable

What is in `gh2-india`, how to run it, and which data still has to come in from outside.

A model nobody else can run is an opinion. Every figure and table in this book is produced by one command, from code anyone can read. The repository is called `gh2-india` and lives at `https://github.com/pritam8abstract/gh2-india` once pushed.

```
gh2-india/
  README.md                                  what each module answers, how to run it
  src/gh2/
    params.py                                every assumption, tagged REPORTED / ESTIMATED / PROJECTED
    lcoh.py                                  levelised cost, five stacks, 100,000 draws
    siteselect.py                            AHP, consistency ratio, weight-flip test
    schedule.py                              CPM, Beta-PERT Monte Carlo, criticality, early release
    resource.py, dispatch.py                 hourly profiles; dispatch LP (HiGHS)
    demand.py                                Bass diffusion demand forecast
    finance.py                               SPV waterfall, DSCR/LLCR/IRR, MMRA, lever ladder
    realoptions.py                           LSMC expansion option; AEM switch
    auctions.py                              the research paper's three tests
    prices.py                                Schwartz one-factor price model
    charts.py                                house-style figures
  scripts/run_all.py                         one command: results/ and figures/
  data/raw/seci_mode2a_tranche1_bids.csv     all 65 bids
  tests/                                     8 sanity tests
  dossiers/                                  LaTeX source of Dossiers 1, 2 and 3
```

9.1 How to run it

On our Windows laptops, in PowerShell:

```
git clone https://github.com/pritam8abstract/gh2-india.git
cd gh2-india
python -m venv .venv; .\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
python scripts/run_all.py                                  about 15 seconds
pytest -q
```

The random seed is fixed (20261013), so every run gives the same numbers. To change an assumption, edit `params.py` and rerun; the figures in `figures/` and the numbers in `results/results.json` update together.

9.2 The data plan

Our build environment could not reach most public data sites, so some feeds are written as functions to run on our own machines. The table is honest about which numbers are real data and which are stand-ins.

DATA	STATUS	USED FOR
SECI Mode 2A bids, 65 rows [?]	In the repo	Research paper, all three tests
PVGIS hourly solar [?]	Function ready	Replaces synthetic solar in the dispatch LP
NASA POWER hourly wind [?]	Function ready	Replaces synthetic wind
CEA grid emission factor [?]	Number in text	Green-certification limit on grid power
World Bank Pink Sheet (urea, gas)	To pull	Fitting the price model
India ammonia import values (HS 2814)	To pull	Grey-ammonia benchmark
PPAC refinery throughput, monthly	To pull	Refinery hydrogen demand base; seasonal forecast
IEX day-ahead prices	To pull	Firm-power scenarios
RBI / FBIL rates and INR–USD	To pull	Debt cost and FX ranges

WHY THIS MATTERS FOR THE PRESENTATION

When Prof. Gopal asks "where does this number come from?", the answer is never "we assumed it". It is "`params.py`, line so-and-so, sourced from this tender result, and here is what happens if it is wrong". That is the difference between a project report and a project book.

The storyline for Prof. Gopal's class

Fourteen slides, one message each, split between two speakers, and the questions we expect.

A good deck is an argument, not a table of contents. Read only the headlines below in order and you should hear the whole case. Each slide has one chart from this book and one sentence the audience should remember.

#	HEADLINE (THE SLIDE TITLE)	EVIDENCE ON THE SLIDE	WHO
1	Green hydrogen is a finance problem wearing an engineering costume	Title; the four numbers (₹522, ₹397, ₹389, 57%)	DS
2	India already uses 6 Mt of hydrogen; almost all of it is grey	Dossier 1: where hydrogen hides; 55–70 Mt of CO_2	SB
3	We chose a hybrid: alkaline for cost, PEM for flexibility, AEM as an option	LCOH table by stack	SB
4	Sell hydrogen to a refinery, not ammonia to an auction	₹279–397/kg refinery contracts vs. ₹53/kg ammonia bids	DS
5	Paradip wins because the buyer is next door, by a whisker	AHP chart; 30.8% flip point	SB
6	The plan says 44 months; the odds say 48	Gantt; merge-bias S-curve	SB
7	The schedule is a finance problem: a 46% chance of beating the waiver deadline	Early release lifts it to 82%	DS
8	Operations: oversize the renewables, hold a 7-hour buffer, buy certified back-up	Dispatch chart; 5.6% grid ceiling	SB
9	As first designed, the plant cannot pay its lender	Base vs. target table; DSCR 0.63	DS
10	Two levers close most of the gap: the stack price and the power price	Lever ladder	DS
11	Bankable on paper, fragile in practice	Monte Carlo IRR; 57% → 32% with an 18-year loan	DS
12	Risk sits in the money, not the machine	Heat map; bow-tie	SB
13	Keep the future cheap: phase two and AEM as options	Option values; stage-gates	SB
14	What we learned, and the paper it became	Auction results; research question	DS

DS: Dhritabrata Swarnakar. SB: Shalvi Bhatnagar. Target length 15 minutes, roughly a minute a slide, with slide 11 given two.

10.1 Mapping to what the course asks for

COURSE THEME	WHERE IT APPEARS
Feasibility	Technical (Dossier 1), market (demand forecast), financial (Section ??), schedule (Section ??)
Project management	Charter, scope, WBS, CPM, PERT, Monte Carlo schedule, earned value, stage-gates
Operations management	AHP site selection, aggregate planning and buffer inventory (dispatch LP), maintenance policy, Kraljic sourcing, demand forecasting
Risk	Register, heat map, bow-tie, contingency at P80, real options
Cost	LCOH, capital budget, lever ladder, required tariff
Stakeholders	Power–interest grid, engagement table, contract map

10.2 The questions we expect, and our answers

"Why not just go alkaline, if it is cheapest?" Alkaline is cheapest at steady load; our power is not steady. The PEM slice takes the fast swings and costs about US$0.19/kg extra. It also buys us the AEM option. If alkaline proves it can ramp without extra wear, we would shift the mix.

"Your schedule is 44 months but you keep quoting 48. Which is it?" 44 is the plan if every task takes its most likely time. Because several paths merge, the median outcome is 48. We promise the lender P80, month 51, and buy three months back by ordering early.

"Isn't ₹397/kg a one-off?" It is one contract, which is why we test ₹279 too, and why we build nothing without a signed buyer. The point of the lever ladder is to show what must be true to live with either price.

"Where do your probabilities come from?" From ranges on each input, every one listed in `params.py` with its source. The ranges are judgements; the method is standard. Change a range and rerun: the code is public.

"What would make you cancel the project?" Failing gate G1: no signed take-or-pay, or an EPC price above US$850/kW, or a P80 date after 2030 that we cannot buy back.

"Is the research paper original?" To our knowledge, nobody has tested the SECI Mode 2A awards against a probabilistic cost of delivery. The early results show that only best-in-class developers can cover cost, and three awards look thin even for them.

ANNEXES

The findings, the inputs and the words

Annex A collects every key finding with its confidence; Annex B lists the main model inputs; Annex C explains the finance terms in plain words.

Annex A • Key findings and how sure we are

#	FINDING	CONFIDENCE
1	The plant should sell hydrogen to a refinery on a long fixed price; ammonia auctions price below most developers' cost.	HIGH
2	Paradip beats Kandla for a hydrogen product, but a 2.5-point change in one weight flips it.	MEDIUM
3	Merge bias adds about 4 months: P50 48.3 vs. a 44-month plan.	MEDIUM
4	Transformers, not electrolysers, are the most likely long pole (critical 52% of the time).	MEDIUM
5	Only a 46% chance of commissioning before the waiver deadline; 82% with a three-month early order release.	MEDIUM
6	Grid power can be at most about 5.6% of supply without breaking the green standard.	HIGH
7	The base design needs ₹522/kg; it fails at ₹397.	HIGH (fails), MEDIUM (level)
8	Stack price and power price close ₹120 of the ₹152/kg gap.	MEDIUM
9	Even the target design breaches a 1.20 cover at least once in 57% of futures; an 18-year loan cuts that to 32%.	MEDIUM
10	The right to expand is worth far more than expanding now; the 2030 cliff is worth about ₹100 crore of preparation.	LOW (level), MEDIUM (logic)
11	Keep the bay AEM-ready; let field durability, not price, trigger the switch.	MEDIUM
12	The 2025 ammonia auctions are coverable only by best-in-class developers; three awards imply returns below 12%.	MEDIUM
13	Green hydrogen demand in India is unlikely to approach 5 Mt by 2030.	LOW (level), MEDIUM (direction)

Annex B • Main model inputs

INPUT	LOW / BASE / HIGH	SOURCE AND TAG
Electrolyser system, hybrid (US$/kW)	900 / 1,250 / 1,750	Dossier 1 market scan ESTIMATED; target 850
Energy use, hybrid (kWh/kg)	50.5 / 53.2 / 57.2	OEM ranges ESTIMATED
Stack life, hybrid (hours)	55,000 / 72,000 / 88,000	ESTIMATED
Delivered power (US$/MWh)	32 / 38 / 48	[?, ?] REPORTED to ESTIMATED

INPUT	LOW / BASE / HIGH	SOURCE AND TAG
Share of hours powered	55% / 62% / 68%	ESTIMATED
Discount rate (LCOH)	8.5% / 10% / 12.5%	ESTIMATED
Debt	70%, 9%, 15 years	ESTIMATED
Tax rate	25.17%	Indian corporate rate REPORTED
O&M	2.1% of electrolyser capex, +4%/yr	ESTIMATED
Degradation	0.6% per year	ESTIMATED
Hydrogen handling (US$m)	15 / 22 / 32	ESTIMATED; target 18
Construction overrun	lognormal, median +5%, P80 +16%	[?] for the skew ESTIMATED
Refinery price (₹/kg)	279 and 397	[?, ?] REPORTED
SIGHT Mode 2A incentive (₹/kg NH_3)	8.82, 7.06, 5.30	[?] REPORTED
Exchange rate (₹/US$)	90 (project), 87 (2025 auctions)	ESTIMATED
Grid emission factor	0.675 kg CO_2/kWh	[?] REPORTED

Annex C • Finance words in plain language

SPV A company created only to own this one plant, so its debts cannot reach the sponsors' other assets.

EBITDA What the plant earns before paying interest, tax and the wear on its machines.

CFADS Cash flow available for debt service: the cash actually left to pay the bank each year.

DSCR CFADS divided by that year's interest plus repayment. 1.30 means thirty paise spare per rupee owed.

LLCR The same idea over the whole life of the loan, in present-value terms.

IRR The yearly return that makes all the money in equal to all the money out. Project IRR counts all the capital; equity IRR counts only the owners' share.

IDC Interest during construction: interest that piles up before the plant earns anything, and gets added to the loan.

MMRA / DSRA Bank accounts the lenders require: one saves for the stack swap, one holds six months of debt payments.

P50 / P80 The value that half, or 80%, of simulated outcomes come in below (or, for schedules, finish within).

Real option The right, but not the duty, to do something later, such as expand or switch technology. It has value because we can wait to see how the world turns out.

Winner's curse In an auction where everyone guesses the cost, the winner is usually the one who guessed lowest, and so is the one most likely to be wrong.

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APA style. Every bracketed number in the text jumps here; every entry links to the original page where one exists. Accessed October 2026.

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